

An Interpretable Machine Learning Framework for Early Customer Churn Detection

Based on the uploaded project: Customer Churn Prediction Using Elastic Net Regularized Logistic Regression

Abstract

Early detection of customer churn is essential for targeted retention. This paper develops an interpretable machine learning framework centered on regularized logistic regression. The framework treats churn prediction as a binary classification problem and emphasizes the explanation of risk factors rather than prediction alone. Customer attributes such as contract characteristics, monthly charges and tenure are incorporated into the modeling process. Preprocessing, regularization, cross-validation and classification metrics form the core of the proposed framework. The objective is to produce actionable churn scores that can be understood by analysts and retention teams.

Introduction

Businesses in telecommunications, banking and e-commerce need to identify customers at risk of leaving. The source project highlights three recurring challenges: high-dimensional and correlated features, overfitting, and the trade-off between accuracy and interpretability. An interpretable model is valuable because stakeholders need to understand not only who is likely to churn, but also why.

Proposed Framework

The proposed framework consists of five stages: data preprocessing, dataset splitting, hyperparameter tuning, model fitting and evaluation. Categorical attributes are encoded and numerical variables standardized. Elastic Net logistic regression is trained after selecting regularization parameters through cross-validation. Coefficients are examined to identify influential customer characteristics. Model performance is evaluated using accuracy, precision, recall, F1-score and ROC-AUC.

Interpretation of Churn Drivers

The source project identifies contract type, monthly charges and customer tenure as leading churn drivers under Elastic Net feature selection. These variables can be transformed into business actions: customers with risky contract structures may receive targeted offers, customers with high monthly charges may be considered for plan optimization, and short-tenure customers may receive stronger onboarding and engagement interventions.

Conclusion

An interpretable churn framework can connect predictive analytics with practical retention decisions. Regularized logistic regression is particularly useful when organizations require transparent model behavior. The proposed framework provides a foundation for real-time customer risk scoring while preserving understandable relationships between customer characteristics and predicted churn.

References / Source Basis

1. Uploaded project presentation, "Customer Churn Prediction Using Elastic Net Regularized Logistic Regression," A.Y. 2026–27. The presentation describes the business problem, Elastic Net rationale, methodology pipeline, evaluation metrics, Telco Customer Churn dataset and related literature.

2. Arivazhagan, B. & Sankara Subramanian, R.S. "Customer Churn Prediction using Logistic Regression with Regularization and Optimization Technique." IJITEE, Vol. 9, Issue 9, 2020.
3. Šimovič, P. P., Chen, C. Y. T. & Sun, E. W. "Classifying the Variety of Customers' Online Engagement for Churn Prediction with a Mixed-Penalty Logistic Regression." Computational Economics, 2023.
4. Yang, L., Li, D. & Lu, Y. "Prediction Modeling and Analysis for Telecom Customer Churn in Two Months." arXiv:1911.00558, 2019.